

Bài 1:

a. $[0, 1]$

$$f(0) = -6 < 0$$

$$f(1) = 2$$

$$\Rightarrow f(0) \cdot f(1) = -12 < 0$$

$$\Rightarrow \text{Có } N^0 \in (0, 1)$$

$$\text{AOCR: } n \geq \log_2 \frac{1-0}{10^{-2}} = \log_2(100)$$

$$= 6,64$$

$\Rightarrow$ cần 7 bước lặp

$$\Rightarrow \text{lần 1: Có: } \frac{0+1}{2} = 0,5$$

$$f(0,5) = -0,625 < 0$$

$$\Rightarrow \text{Nghiem nằm trong } [0,5; 1]$$

$$\Rightarrow \text{lần 2: } \frac{0,5+1}{2} = 0,75$$

$$f(0,75) = 0,984 > 0$$

$$\Rightarrow N^0 \in [0,5; 0,75]$$

$$\Rightarrow \text{lần 3: Có: } \frac{0,5+0,75}{2} = 0,625$$

$$f(0,625) = 0,258 > 0$$

$$\Rightarrow N^0 \in [0,5; 0,625]$$

$$\Rightarrow \text{lần 4: } f(0,5625) = -0,162 < 0$$

$$\Rightarrow N^0 \in [0,5625; 0,625]$$

$$\Rightarrow \text{lần 5: } \frac{0,5625+0,625}{2} = 0,59375$$

$$f(0,59375) = 0,054 > 0$$

$$\Rightarrow N^0 \in [0,5625; 0,59375]$$

$$\Rightarrow \text{lần 6: Có: } \frac{0,5625+0,59375}{2}$$

$$= 0,578$$

$$f(0,578) = 0,05 < 0$$

$$\Rightarrow N^0 \in [0,578; 0,59375]$$

$\Rightarrow$ lần 7:

$$\text{Có: } \frac{0,578+0,59375}{2} = 0,5858$$

Vậy đến 7 bước, N^0 gần đúng,

$$x \approx 0,586, \text{ sai số } 10^{-2}$$

b. $[1; 3; 2]$

$$\text{Có: } f(1) = 2 > 0$$

$$f(3,2) = -0,112 < 0$$

$$\Rightarrow \text{Có } N^0 \text{ trong } (1; 3,2)$$

$$\text{AOCR: } n \geq \log_2 \frac{3,2-1}{10^{-2}} \approx 7,78$$

Cần 8 bước lặp

$$\Rightarrow \text{lần 1: } \frac{1+3,2}{2} = 2,1$$

$$f(2,1) = 0,231 > 0$$

$$\Rightarrow N^0 \in [2,1; 3,2]$$

$$\Rightarrow \text{lần 2: } \frac{2,1+3,2}{2} = 2,65$$

$$f(2,65) = -0,536 < 0$$

$$\Rightarrow N^0 \in [2,1; 2,65]$$

$$\Rightarrow \text{lần 3: } \frac{2,1+2,65}{2} = 2,375$$

$$f(2,375) = -0,334 < 0$$

$$\Rightarrow N^0 \in [2,1; 2,375]$$

$$\Rightarrow \text{lần 4: } \frac{2,1+2,375}{2} = 2,2375$$

$$f(2,2375) = -0,082 < 0$$

$$\Rightarrow N^0 \in [2,1; 2,2375]$$

$$\Rightarrow \text{lần 5: } \frac{2,1+2,2375}{2} = 2,16875$$

$$f(2,16875) = 0,05 > 0$$

$$\Rightarrow N^0 \in [2,16875; 2,2375]$$

$$\Rightarrow \text{Lan 6: } \frac{2,16875 + 2,2375}{2} = 2,203125$$

$$f(2,203125) = -0,015 < 0$$

$$\Rightarrow N^0 \in [2,16875; 2,203125]$$

$$\Rightarrow \text{Lan 7:}$$

$$C6: \frac{2,16875 + 2,203125}{2} = 2,1859375$$

$$f(2,1859375) = 0,010 > 0$$

$$\Rightarrow N^0 \in [2,1859375; 2,203125]$$

$$\Rightarrow \text{Lan 8:}$$

$$C6: \frac{2,1859375 + 2,203125}{2} = 2,19453125$$

$$\text{Vậy } x \approx 2,195 \text{ với sai số } \leq 10^{-2}$$

$$C. [3,2; 4]$$

$$C6: f(3,2) = -0,112 < 0$$

$$f(4) = 2 > 0$$

$$\Rightarrow C6 N^0 \in (3,2; 4)$$

$$\text{ANCT } n \geq \log_2(80) = 6,32$$

$$\Rightarrow \text{Cần 7 bước lặp.}$$

$$\Rightarrow \text{L1: } \frac{3,2 + 4}{2} = 3,6$$

$$f(3,6) = -0,384 < 0$$

$$\Rightarrow N^1 \in [3,6; 4]$$

$$\Rightarrow \text{L2: } \frac{3,6 + 4}{2} = 3,8$$

$$f(3,8) = 0,312 > 0$$

$$\Rightarrow N^2 \in [3,6; 3,8]$$

$$\Rightarrow \text{L3: } \frac{3,6 + 3,8}{2} = 3,7$$

$$f(3,7) = -0,107 < 0$$

$$\Rightarrow N^3 \in [3,7; 3,8]$$

$$\Rightarrow \text{L4: } \frac{3,7 + 3,8}{2} = 3,75$$

$$f(3,75) = 0,070 > 0$$

$$\Rightarrow N^4 \in [3,7; 3,75]$$

$$\Rightarrow \text{L5: } C6: \frac{3,7 + 3,75}{2} = 3,725$$

$$f(3,725) = -0,022 < 0$$

$$\Rightarrow N^0 \in [3,725; 3,75]$$

$$\Rightarrow \text{Lan 6: } \frac{3,725 + 3,75}{2} = 3,7375$$

$$f(3,7375) = 0,024 > 0$$

$$\Rightarrow N^0 \in [3,725; 3,7375]$$

$$\Rightarrow \text{Lan 7: } \frac{3,725 + 3,7375}{2} = 3,73125$$

$$\text{Vậy } x \approx 3,73 \text{ với sai số } \leq 10^{-2}$$

Bài 2:

$$a [-2; -1]$$

$$C6: f(-2) = 12 > 0$$

$$f(-1) = -1 < 0$$

$$\Rightarrow f(-2) \cdot f(-1) = -12 < 0$$

$$\Rightarrow \text{pt có } N^0 \in (-2; -1)$$

$$\text{ANCT. } n \geq \frac{\log_2(-1) - \log_2(-2)}{10^{-2}}$$

$$\approx 6,64$$

Tạo bảng sau:

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	-2	-1	-1,5	13,16 > 0
2	-1,5	-1	-1,25	$\frac{23,1}{256} < 0$
3	-1,5	-1,25	-1,375	$\frac{-1163}{4096} < 0$
4	-1,5	-1,375	-1,4375	$\frac{1,195705}{2^{13}} > 0$
5	-1,4375	-1,375	-1,40625	$-0,027 < 0$
6	-1,4375	-1,40625	-1,421875	0,0626 > 0
7	-1,421875	-1,40625	-1,4140625	

$$\text{Vậy } x \approx -1,42$$

$$b, [0, 2]$$

$$f(0) = 4 > 0$$

$$f(2) = -4 < 0$$

$$\Rightarrow f(0) \cdot f(2) = -16 < 0$$

$$\Rightarrow n^0 \in (0, 2)$$

$$AOCF: \Rightarrow n > \log_2 200 \approx 7,64$$

$$\Rightarrow n \geq 8$$

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	0	2	1	3 > 0
2	1	2	1,5	$-\frac{11}{16} < 0$
3	1	1,5	1,25	$\frac{329}{256} > 0$
4	1,25	1,5	1,375	$\frac{1281}{4096} > 0$
5	1,375	1,5	1,4375	$-0,186... < 0$
6	1,375	1,4375	1,40625	$0,063... > 0$
7	1,40625	1,4375	1,421875	$-0,0613 < 0$
8	1,40625	1,421875	1,4140625	

$$\Rightarrow x \approx 1,414$$

$$c, [2, 3]$$

$$f(2) = -4 < 0$$

$$f(3) = 7 > 0$$

$$\Rightarrow f(2) \cdot f(3) < 0 \Rightarrow n^0 \in (2, 3)$$

$$AOCF: n > \log_2 100 \Rightarrow n \geq 7$$

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	2	3	2,5	$-\frac{81}{16} < 0$
2	2,5	3	2,75	$\frac{89}{256} < 0$
3	2,5	2,75	2,625	$-1,357 < 0$
4	2,625	2,75	2,6875	$-0,7956 < 0$
5	2,6875	2,75	2,71875	$-0,2474 < 0$

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
6	2,71875	2,75	2,734375	$0,04412 > 0$
7	2,71875	2,734375	2,7265625	

$$\Rightarrow x \approx 2,73$$

$$d: [-1, 0]$$

$$f(-1) = -1 < 0$$

$$f(0) = -4 > 0$$

$$\Rightarrow f(-1) \cdot f(0) < 0 \Rightarrow n^0 \in (-1, 0)$$

$$AOCF: n > \log_2 100 \Rightarrow n \geq 7$$

$$n^0 \in (-1, 0) \Rightarrow n \geq 7$$

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	-1	0	-0,5	$\frac{21}{16} > 0$
2	-1	-0,5	-0,75	$-\frac{21}{156} < 0$
3	-0,75	-0,5	-0,625	$\frac{2369}{4096} > 0$
4	-0,75	-0,625	-0,6875	$0,232... > 0$
5	-0,75	-0,6875	-0,71875	$0,680... > 0$
6	-0,75	-0,71875	-0,734375	$-0,0117... < 0$
7	-0,734375	-0,71875	-0,7265625	

$$\text{Vậy } x \approx -0,73$$

Bài 3

$$a. x - 2^{-x} = 0 \text{ với } 0 \leq x \leq 1$$

$$f(0) = -1 < 0 \Rightarrow n^0 \in (0, 1)$$

$$f(1) = 0,5 > 0$$

$$\Rightarrow n > \log_2 \frac{1}{10,5} \approx 16,6 \Rightarrow n \geq 17$$

$$\Rightarrow n \geq 17$$

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	0	1	0,5	$-0,2071$
2	0,5	1	0,75	$0,155...$
3	0,5	0,75	0,625	$-0,0237$
4	0,625	0,75	0,6875	$0,0666$
5	0,625	0,6875	0,65625	$0,0219$
6	0,625	0,65625	0,640625	$-0,0009$

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
7	0,640625	0,65625	0,6484375	0,0105
8	0,640625	0,6484375	0,64453125	0,0098
9	0,640625	0,64453125	0,642578125	0,0019
10	0,640625	0,642578125	0,6416015625	0,0005
11	0,640625	0,6416015625	0,641113281	-0,0002
12	0,641113281	0,6416015625	0,641357422	0,00015
13	0,641113281	0,641357422	0,641235352	-0,00002
14	0,641235352	0,641357422	0,641296387	0,00006
15	0,641235352	0,641296387	0,641265870	0,00027
16	0,641235352	0,641265870	0,641250610	0,000339
17	0,641235352	0,641250610	0,641242981	

Var $x \approx 0,64124$.

b. $f(0) = -1 < 0 \Rightarrow f(0) \cdot f(1) < 0$
 $f(1) = 2 > 0 \Rightarrow$ c6 no $\in (0,1)$

c. $n > \log_2 \frac{1}{10^5} = 16,6 \Rightarrow n > 17$

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	0	1	0,5	-0,10128
2	0,5	1	0,75	0,72237
3	0,5	0,75	0,625	0,28867
4	0,5	0,625	0,5625	0,0896
5	0,5	0,5625	0,53125	-0,00902
6	0,53125	0,5625	0,546875	0,03873
7	0,53125	0,546875	0,5390625	0,01477
8	0,53125	0,5390625	0,53515625	0,00283
9	0,53125	0,53515625	0,532203125	-0,00310
10	0,532203125	0,53515625	0,534179688	-0,00013
11	0,534179688	0,53515625	0,53667369	0,00133
12	0,534179688	0,53667369	0,53467669	0,00061
13	0,534179688	0,53467669	0,53442385	0,00024
14	0,534179688	0,53442385	0,53430125	0,00006
15	0,534179688	0,53430125	0,53424023	0,00003
16	0,534179688	0,53424023	0,53425464	0,00001

17. $0,534210703$
 $\Rightarrow x \approx 0,53422$
c. $2x \cos(x) - (x+1)^2 = 0$
1. $-3 \leq x \leq -2$
2. $-1 \leq x \leq 0$
 $f(-3) < 0 \Rightarrow f(-3) \cdot f(-2) < 0$
 $f(-2) < 0 \Rightarrow$ c6 no $\in (-3, -2)$
 $\Rightarrow n > 17$

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	-3	-2	-2,5	76
2	-3	-2,5	-2,75	<0
3	-2,75	-2,625	-2,6875	>0
4	-2,75	-2,625	-2,6875	<0
5	-2,6875	-2,625	-2,65625	<0
6	-2,65625	-2,625	-2,640625	>0
7	-2,640625	-2,625	-2,633125	>0
8	-2,640625	-2,633125	-2,636875	>0
9	-2,640625	-2,636875	-2,63875	>0
10	-2,640625	-2,63875	-2,6396875	>0
11	-2,640625	-2,6396875	-2,6396875	>0
12	-2,640625	-2,6396875	-2,6396875	>0
13	-2,6396875	-2,6396875	-2,6396875	>0
14	-2,6396875	-2,6396875	-2,6396875	>0
15	-2,6396875	-2,6396875	-2,6396875	>0
16	-2,6396875	-2,6396875	-2,6396875	>0
17	-2,6396875	-2,6396875	-2,6396875	>0

$\Rightarrow x \approx -2,63973$

2. $f(-1) > 0, f(0) < 0$
 $\Rightarrow f(1) \cdot f(0) < 0$
 $\Rightarrow$ c6 no $\in (-1, 0)$
 $\Rightarrow n > \log_2 10^5 \approx 17$

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	-1	0	-0.5	<0
2	-1	-0.5	-0.75	<0
3	-1	-0.75	-0.875	<0
4	-0.875	-0.75	-0.8125	>0
5	-0.8125	-0.75	-0.78125	>0
6	-0.78125	-0.75	-0.765625	<0
7	-0.765625	-0.75	-0.7578125	<0
8	-0.7578125	-0.75	-0.75390625	<0
9	-0.75390625	-0.75	-0.751953125	<0
10	-0.751953125	-0.75	-0.7509765625	<0
11	-0.7509765625	-0.75	-0.75048828125	<0
12	-0.75048828125	-0.75	-0.750244140625	<0
13	-0.750244140625	-0.75	-0.7501220703125	<0
14	-0.7501220703125	-0.75	-0.75006103515625	<0
15	-0.75006103515625	-0.75	-0.750030517578125	<0
16	-0.750030517578125	-0.75	-0.7500152587890625	<0
17	-0.7500152587890625	-0.75	-0.75000762939453125	<0

ADCT: $n \geq \log_2 \frac{1}{10^{-5}} \Rightarrow n \geq 17$

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	1	2	1.5	<0
2	1	1.5	1.25	>0
3	1.25	1.5	1.375	>0
4	1.375	1.5	1.4375	>0
5	1.4375	1.5	1.46875	>0
6	1.46875	1.5	1.484375	>0
7	1.484375	1.5	1.4921875	<0
8	1.4921875	1.5	1.49709375	<0
9	1.49709375	1.5	1.499546875	<0
10	1.499546875	1.5	1.4998734375	<0
11	1.4998734375	1.5	1.49993671875	<0
12	1.49993671875	1.5	1.499968359375	<0
13	1.499968359375	1.5	1.4999841796875	<0
14	1.4999841796875	1.5	1.49999208984375	<0
15	1.49999208984375	1.5	1.499996044921875	<0
16	1.499996044921875	1.5	1.4999980224609375	<0
17	1.4999980224609375	1.5	1.49999901123046875	<0

d. $f(0.2) < 0 \Rightarrow f(0.1) \cdot f(0.3) < 0$
 $f(0.3) > 0 \Rightarrow (0.1, 0.3)$
 $n \geq \log_2 \frac{0.1}{10^{-5}} \Rightarrow n \geq 17$

b. $f(0) = 2 > 0$
 $f(1) = 2 + 3 \cos 1 - e > 0$
 $\Rightarrow f(0) \cdot f(1) > 0$
 $\Rightarrow n \geq 17$

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	0.2	0.3	0.25	<0
2	0.2	0.25	0.225	<0
3	0.225	0.25	0.2375	<0
4	0.2375	0.25	0.24375	<0
5	0.24375	0.25	0.246875	<0
6	0.246875	0.25	0.2484375	<0
7	0.2484375	0.25	0.24921875	<0
8	0.24921875	0.25	0.249609375	<0
9	0.249609375	0.25	0.2498046875	<0
10	0.2498046875	0.25	0.24990234375	<0
11	0.24990234375	0.25	0.249951171875	<0
12	0.249951171875	0.25	0.2499755859375	<0
13	0.2499755859375	0.25	0.24998779296875	<0
14	0.24998779296875	0.25	0.249993896484375	<0

c. $1 \leq x \leq 2$
 $f(1) \cdot f(2) < 0$
 $\Rightarrow n \geq 17$

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	1	2	1.5	<0
2	1	1.5	1.25	<0
3	1.25	1.5	1.375	<0
4	1.375	1.5	1.4375	<0
5	1.4375	1.5	1.46875	<0
6	1.46875	1.5	1.484375	<0
7	1.484375	1.5	1.4921875	<0
8	1.4921875	1.5	1.49709375	<0
9	1.49709375	1.5	1.499546875	<0
10	1.499546875	1.5	1.4998734375	<0
11	1.4998734375	1.5	1.49993671875	<0
12	1.49993671875	1.5	1.499968359375	<0
13	1.499968359375	1.5	1.4999841796875	<0
14	1.4999841796875	1.5	1.49999208984375	<0
15	1.49999208984375	1.5	1.499996044921875	<0
16	1.499996044921875	1.5	1.4999980224609375	<0
17	1.4999980224609375	1.5	1.49999901123046875	<0

$\Rightarrow x \approx 0.23733$
 Bài 4. $e \leq 10^{-5}$
 $a \cdot 3x - e^x = 0 \quad \forall x \in [1, 2]$
 $f(1) = 3 - e$
 $f(2) = 6 - e^2$
 $\Rightarrow f(1) \cdot f(2) = 6e - 3e^2 - 6e + e^3$
 $= e^3 - 3e^2 < 0$

Bài 5:

$$x^2 - 2 = \cos(x^2 - 2), \quad \varepsilon \leq 10^{-5}$$

trên $[0.5; 1.5]$

$$C6: x^2 - 2 = \cos(x^2 - 2)$$

$$\Leftrightarrow x^2 - 2 - \cos(x^2 - 2) = 0$$

$$f(x) = x^2 - 2 - \cos(x^2 - 2)$$

$$\Rightarrow f(0.5) = -1.351 < 0$$

$$f(1.5) = 1.983 > 0$$

$$\Rightarrow f(0.5) \cdot f(1.5) < 0$$

$$\Rightarrow C6 \text{ no } \in (0.5; 1.5)$$

$$ADCF: n \geq 17 \log_2 \frac{1}{10^{-5}} \approx 17$$

Phân hoạch

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	0.5	1.5	1	<0
2	1	1.5	1.25	>0
3	1	1.25	1.125	>0
4	1	1.125	1.0625	>0
5	1	1.0625	1.03125	<0
6	1.03125	1.0625	1.046875	<0
7	1.046875	1.0625	1.054688	<0
8	1.054688	1.0625	1.058599	>0
9	1.054688	1.058599	1.055641	>0
10	1.054688	1.055641	1.055176	<0
11	1.055176	1.055641	1.055410	>0
12	1.055176	1.055410	1.055298	<0
13	1.055298	1.055410	1.055359	<0
14	1.055359	1.055410	1.055374	>0
15	1.055374	1.055410	1.055367	>0
16	1.055367	1.055410	1.055367	<0
17	1.055367	1.055367	1.055367	<0

$$\Rightarrow x \approx 1.05537$$

Bài 6:

GIS chọn $[0.5; 2.4]$ là 1 đoạn hội tụ của $f(x)$

$$C6: f(0.5) < 0$$

$$f(2.4) > 0$$

Phân hoạch sau.

n	a	b	$\frac{a+b}{2}$	$f(\frac{a+b}{2})$
1	0.5	2.4	0.95	<0
2	0.95	2.4	1.675	>0
3	0.95	1.675	1.3125	>0
4	0.95	1.3125	1.13125	>0
5	0.95	1.13125	1.040625	>0
6	0.95	1.040625	0.995311	<0
7	0.995311	1.040625	1.017969	>0
8	0.995311	1.017969	1.006641	>0
9	0.995311	1.006641	1.000977	>0
10	0.995311	1.000977	0.998144	<0

$\Rightarrow$ No. hội tụ

$$x \approx 1$$

Phân hoạch số do 1 lần no hội tụ